A Multi-Agent Orchestration Framework for Venture Capital Due Diligence

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Abstract

We present a fully automated multi-agent framework for corporate due diligence and market analysis in venture capital. The system runs on an event-driven orchestration architecture, combining Large Language Models (LLMs) with real-time web retrieval to synthesize unstructured data into investment intelligence. A core component reverse-engineers the frontend-to-backend communication of the Greek Business Registry (Γ.Ε.ΜΗ.), querying dynamic endpoints and applying Optical Character Recognition (OCR) to extract data from official financial filings and corporate records. When registry data is absent, a structural fallback queries alternative financial databases rather than generating unverified figures. Together, these components show that modular agentic workflows can substantially reduce analyst time in due diligence while preserving data provenance.

Keywords: AI agents; Workflow Automation; Due Diligence; Venture Capital; Financial Intelligence; Large Language Models; Hallucination Mitigation

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1 Introduction

Evaluating prospective investments and monitoring portfolio companies demands synthesis of fragmented information across incompatible sources: unstructured web data, real-time news, competitive analyses, and official financial registries that are often locked behind administrative access controls. Manual desk research, the standard approach, introduces human error, cognitive bias, and delays that compound across a portfolio (Gompers et al., 2016; Kahneman, 2011).

LLMs offer new capabilities for natural language understanding and data synthesis (Bommasani et al., 2021), but deploying them for financial due diligence has three structural problems. Training data cut-offs leave models blind to current market signals. Models generate plausible but factually wrong information, a failure mode investment decisions cannot tolerate (Ji et al., 2023). And LLMs cannot independently query external databases, call APIs, or parse scanned financial PDFs (Q. Wu et al., 2023).

We address these problems with a fully automated, multi-agent orchestration pipeline. Rather than a single LLM prompt, the system decomposes due diligence into specialized sub-tasks handled by distinct AI agents (Xi et al., 2023). Real-time search APIs supply current market intelligence; custom extraction modules programmatically retrieve and parse official financial filings from the Greek Business Registry (Γ.Ε.ΜΗ.).

This paper makes three contributions: (1) a modular, multi-agent architecture that automates end-to-end corporate research; (2) a programmatic extraction pipeline that queries hidden registry endpoints, retrieves PDF documents, and applies OCR to structure financial data; and (3) a structural fallback mechanism that switches to open-source intelligence when official registry data is unavailable, replacing unverifiable LLM outputs with explicit data gaps (Gao et al., 2023).

2 Background and Related Work

The field has moved from single-prompt LLM interactions toward Agentic Workflows, where multiple LLM-powered entities each carry distinct roles, system prompts, and tool-use capabilities (Xi et al., 2023). Chaining agents lets each one hand its verified output to the next, building reasoning chains too complex for a single model call. Multi-step analytical problems, which require precision and iterative correction, are the natural fit for these architectures (Q. Wu et al., 2023).

Applying these systems to finance is natural: general LLMs struggle with specialized vocabulary and numerical precision (Yang et al., 2023). Instruction tuning combined with real-time data feeds yields more precise risk assessment and market sentiment analysis (S. Wu, Irsoy, et al., 2023). Frameworks such as MARAG-Fin and QuantAgents go further, distributing analytical tasks across specialized agents to improve accuracy and reduce hallucinations in volatile markets (Li et al., 2025; Luckianto & Gunawan, 2026). Investment analysis has become a productive testbed for task-specific agentic pipelines.

Dynamic information retrieval addresses the static knowledge problem: grounding model outputs in real-time search results and external data streams lets systems sidestep training cut-offs entirely (Lewis et al., 2020). For corporate intelligence, Open-Source Intelligence (OSINT) provides current market sizes, funding rounds, and competitor movements. Retrieval-Augmented Generation makes that intelligence verifiable by linking outputs to primary sources (Gao et al., 2023).

Automated OCR and document processing extend this reach into non-digitized corporate records. Layout-aware parsers can handle balance sheets and income statements without the structural loss common to older engines (Paruchuri, 2025). Combined with real-time web retrieval, these tools give workflows a path from fragmented market signals to formal regulatory filings in a single pipeline (Bommasani et al., 2021).

3 System Architecture

The system runs on an event-driven automation platform (n8n, 2024), structured as a Directed Acyclic Graph (DAG) of processing nodes. A custom HTML form acts as the entry point for investment teams: analysts select a target portfolio company and trigger the full research workflow with a single click, with the low-code backend hidden from view.

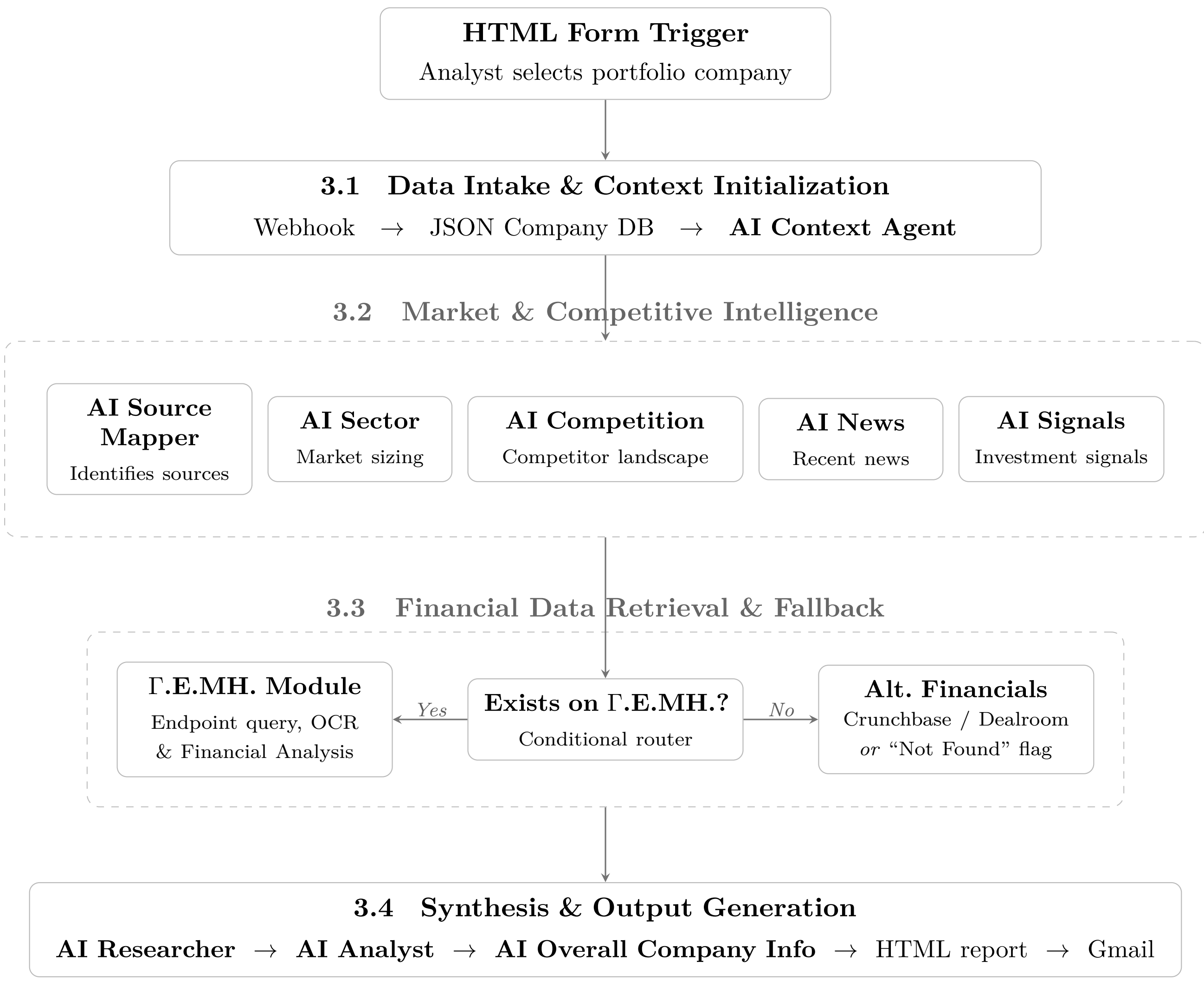

Figure 1: High-level overview of the event-driven multi-agent orchestration architecture, structured according to the four pipeline stages described in Sections 3–3.4.

3.1 Data Intake and Context Initialization

On submission, a Webhook node captures the payload. A transformation node maps the selected company to a pre-scraped JSON database of baseline attributes: founders, sector, initial investment year, headquarters, and registration numbers. The **AI Context Agent** takes this input and produces a compact company profile that anchors all downstream research nodes.

3.2 Market and Competitive Intelligence Modules

Five specialized agents query the Perplexity Sonar Deep Research API (Perplexity AI, 2024) for real-time market intelligence. The **AI Source Mapper** runs first, identifying the most reliable sector-specific portals, databases, and news aggregators for the target company. The **AI Sector** and **AI Competition** agents run in parallel against those sources, extracting market sizing, macro-environmental trends, and a competitive landscape that categorizes competitors as direct, adjacent, or niche innovators. Once both complete, the **AI News** and **AI Signals** agents scan for recent strategic developments (partnerships, product launches) and extract quantifiable investment signals.

3.3 Financial Data Retrieval and Fallback Mechanisms

The most complex component interfaces with the Greek Business Registry (Γ.Ε.ΜΗ.). Although the registry provides an official API, obtaining authenticated credentials carries administrative overhead. Instead, the framework reverse-engineers the platform's frontend-to-backend communication, a method validated by prior open-source work targeting the same registry (Drakakis, 2025).

Registry data and corporate announcements are served as PDF documents, loaded by the public portal from specific backend endpoints. A conditional router (Switch/If nodes) checks for a valid registry number (arGEMI). When one exists, an HTTP Request node mimics the browser's behavior, querying those endpoints directly to retrieve the JSON payload of required document IDs.

Document IDs are sorted into two streams: **Corporate Modifications (Announcements)**, covering board changes, capital increases, and statutory modifications; and **Financial Statements**, comprising official balance sheets, income statements, and auditor reports.

The pipeline fetches raw PDF files (prioritizing the most recent fiscal years) and runs them through a layout-aware text extractor (Paruchuri, 2025). Two summarization models, **AI FinSummary** and **AI ModSummary**, then distill the extracted text into standardized financial metrics (Assets, Liabilities, Revenue, EBIT) with source citations preserved.

Preventing fabricated financial data is the system's hardest constraint. When the initial routing finds no Γ.Ε.ΜΗ. number (typically companies incorporated abroad), the pipeline skips the registry branch and activates the **Alternative Financials** agent, which queries Crunchbase, Dealroom, and similar open-source databases. If those sources return nothing, the agent writes a "Not Found" flag directly into the report, making data gaps explicit rather than filling them with generated figures.

3.4 Synthesis and Output Generation

Three synthesis agents combine the intelligence streams. The **AI Researcher** aggregates sector, competition, news, signals, and financial data into a strategic research note that surfaces recent developments and blind spots. The **AI Analyst** takes that note and produces an Executive Summary, assigns attractiveness scores across market timing and product differentiation, and drafts 30-to-180-day recommendations for both the fund and the startup. The **AI Overall Company Info** agent generates a public-facing summary founders can use to understand their external market footprint.

A formatting node converts the JSON outputs into a styled HTML report and dispatches it to the requesting analyst through a Gmail API integration.

4 Data and Workflow Availability

All system artifacts are publicly available in a dedicated repository (Alexandrou, 2026): the n8n JSON workflow file, project documentation, and an explanatory video walk-through. Consult the repository documentation for the required environment variables and node configurations before attempting replication.

5 Conclusions and Future Work

Multi-agent orchestration can automate corporate due diligence at a level of reliability previously requiring human analysts. By combining dynamic web retrieval, reverse-engineered endpoint querying, and OCR-based document processing, the n8n pipeline connects unstructured market signals to official financial compliance data in a single automated pass. Structural fallback mechanisms prevent the system from generating unverified financial figures, replacing hallucinated outputs with explicit data gaps.

Three directions extend this work. First, integrating the pipeline with the fund's internal portfolio database would let each report draw on proprietary deal-level data, historical performance metrics, and analyst notes specific to each startup, making the output contextual rather than general. Second, adding international registries such as UK Companies House (UK Companies House, 2025) and equivalent European authorities would extend coverage to companies incorporated outside Greece. Third, replacing the third-party PDF extraction API with a self-hosted open-source alternative would cut operational costs and remove the external dependency without degrading extraction quality.

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